Kosuke Ukita

PhD student (D1), full-time

Kyushu Institute of Technology

ukita.kosuke299@mail.kyutech.jp GitHub Scholar ORCID

EDUCATION

Apr. 2026 - Present

PhD

Kyushu Institute of Technology
Department of Computer Science and Systems Engineering,
Graduate School of Computer Science and Systems Engineering

Thesis: ""

GPA: -- / 4.0

Apr. 2024 - Mar. 2026

MSc

Kyushu Institute of Technology
Department of Creative Informatics,
Graduate School of Computer Science and Systems Engineering

Thesis: "A Study on Sensor Foundation Models via Joint Training using Flow Matching"

GPA: 3.6 / 4.0

Apr. 2020 - Mar. 2024

BSc

Kyushu Institute of Technology
Department of Artificial Intelligence,
Computer Science and Systems Engineering

Thesis: "Proposal of Representation-Conditional Latent Diffusion Models: Hematoma Classification in Brain CT Images"

GPA: 3.7 / 4.0

Apr. 2017 - Mar. 2020

High School Diploma

Kanonji Daiichi High School
Science and Mathematics Course

GPA: 4.9 / 5.0

WORK EXPERIENCE

Dec. 2023 - Mar. 2026

Research Assistant (RA)

Superviser: Tsuyoshi Okita

Researched and Developed foundation models

Apr. 2024 - Jun. 2026

Teaching Assistant (TA)

Lecturer: Tsuyoshi Okita

- Deep Learning（深層学習）
- Deep Learning Exercise（知能情報工学プロジェクト-深層学習演習）
- Deep Learning Advanced I（深層学習特論I-大学院科目）

Oct. 2023 - Nov. 2023

OPTiM Corporation

Research Intern / Part-time Research Assistant

Engaged in research and developed AI technologies for agricultural DX

PUBLICATIONS

[1] **Efficient Inference for Flow Matching via Unified Path CFG**
Kosuke Ukita, Tsuyoshi Okita
The 30th Pacific-Asia Conference on Knowledge Discovery and Data Mining (PAKDD 2026)
June 2026 · Hong Kong, China
Oral refereed

[2] **High-Performance Self-Supervised Learning by Joint Training of Flow Matching**
Kosuke Ukita, Tsuyoshi Okita
The 29th International Conference on Artificial Intelligence and Statistics (AISTATS 2026)
May 2026 · Tangier, Morocco
Spotlight refereed

[3] **Foundation Models to Tackle Activity Recognition in Unknown Domain: Sussex-Huawei Locomotion Challenge 2025 Task 2**
Tsuyoshi Okita, Kosuke Ukita, Asahi Miyazaki, Daichi Kubota, Jukichi Ota, Naoki Kagiyama, Asahi Nishikawa, Nagayasu Daichi, Syunya Tomitaka, Daisuke Nozaki, Yuki Odo, Raku Yamashita, Xiaolong Ye, Huayu Gao, Kazuki Okahashi, Koki Matsuishi, Masaharu Kagiyama, Kodai Hirata, Haruki Kai, Lin Wang, Hristijan Gjoreski, Mathias Ciliberto, Paula Lago, Kazuya Murao, Daniel Roggen
Companion of the 2025 ACM International Joint Conference on Pervasive and Ubiquitous Computing (UbiComp '25), pp. 983–991
October 2025 · Espoo, Finland
Oral | refereed

[4] **Multimodal Foundation Model for Cross-Modal Retrieval and Activity Recognition Tasks**
Koki Matsuishi, Kosuke Ukita, Tsuyoshi Okita
International Journal of Activity and Behavior Computing (IJABC), Care XDX Center, Kyushu Institute of Technology, Vol. 2025, No. 3, pp. 1-25
November 2025 · Kitakyushu, Japan
Oral | refereed

[5] **Energy-Efficient High-Performance Representation Learning for Sensor Data via Flow Matching**
Kosuke Ukita, Tsuyoshi Okita
The 7th International Symposium on Neuromorphic AI Hardware, Paper No. P-19
October 2025 · Kitakyushu, Japan
Poster | non-refereed

[6] **Image Classification Using a Diffusion Model as a PreTraining Model**
Kosuke Ukita, Xiaolong Ye, Tsuyoshi Okita
arXiv preprint arXiv:2505.06890, pp. 1-10
May 2025
Preprint | non-refereed

[7] **AURA-MFM: 詳細な行動理解のためのCLIPを用いたマルチモーダル基盤モデルの構築**
松石康希, 浮田嵩祐, 大北剛
情報処理学会研究報告(IPSJ SIG Technical Report), Vol. 2025-UBI-85, No. 8
February 2025 · Osaka, Japan
Oral | non-refereed

[8] **Improving Human Activity Recognition through Diffusion Models**
Kosuke Ukita, Tsuyoshi Okita
The International Symposium on Applied Engineering and Sciences (SAES2024), No. p0052
November 2024 · Kitakyushu, Japan
Oral | non-refereed

[9] **Ensemble Learning of Models for Human Activity Recognition**
Asahi Miyazaki, Kosuke Ukita, Tsuyoshi Okita
The International Symposium on Applied Engineering and Sciences (SAES2024), No. p0053
November 2024 · Kitakyushu, Japan
Oral | non-refereed

[10] **Analysis of Human Activity Recognition by Diffusion Models**
Kosuke Ukita, Tsuyoshi Okita
Companion of the 2024 ACM International Joint Conference on Pervasive and Ubiquitous Computing (UbiComp '24), pp. 458–463
October 2024 · Melbourne, Australia
Oral | refereed

[11] **表現条件付き潜在拡散モデルと表現学習**
浮田嵩祐, Ye Xiaolong, 大北剛
情報処理学会研究報告, 火の国情報シンポジウム 2024, No. A4-4
March 2024 · Kagoshima, Japan

`Oral` `non-refereed`

[12] **Towards LLMs for Sensor Data: Multi-Task Self-Supervised Learning**
Tsuyoshi Okita, Kosuke Ukita, Koki Matsuishi, Masaharu Kagiyama, Kodai Hirata, Asahi Miyazaki
Adjunct Proceedings of the 2023 ACM International Joint Conference on Pervasive and Ubiquitous Computing & the 2023 ACM International Symposium on Wearable Computing (UbiComp/ISWC '23 Adjunct), pp. 499–504
October 2023 · Cancun, Mexico

`Oral` `refereed`

GRANTS

– **Kyushu Institute of Technology SPRING Scholarship Awardee**
Japan Science and Technology Agency (JST)
Apr. 2026 - Present · JPY ¥1,800,000 (+ ¥900,000) / year

– **Student Travel Award**
The 30th Pacific-Asia Conference on Knowledge Discovery and Data Mining (PAKDD 2026)
Jun. 2026 · USD $500

– **International Travel Grant**
The Telecommunications Advancement Foundation
Apr. 2026 · JPY ¥290,000

AWARDS

– **Student Travel Award**
The 30th Pacific-Asia Conference on Knowledge Discovery and Data Mining (PAKDD 2026)
Jun. 2026

– **Spotlight Presentation (Top 3%)**
The 29th International Conference on Artificial Intelligence and Statistics (AISTATS 2026)
May. 2026

– **Promising Researcher Award**
Information Processing Society of Japan (IPSJ) Kyushu Branch
Mar. 2024

– **Best Research Award**
The Japan Statistical Society (JSS), Special Committee of Statistical Education (CSE), The 8th Sports Data Analysis Competition
May. 2019

SKILLS

LANGUAGES

Python · C++ · C · Java · JavaScript · TypeScript · Go · SQL · R

FRAMEWORKS & TOOLS

PyTorch · TensorFlow · Hugging Face · Jupyter · ABCI · Docker · Git

OTHER SKILLS

Deep Learning · Neural Networks · Model Training · Data Processing · API Development · Linux